Policy Gradient Methods

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value based → policy based

$\pi(a|s) \to \pi(a|s,\theta)$

Basic idea of policy gradient:

① metric $J(\theta) \xrightarrow{\text{def}}$ optimal policy

② optimization $\theta_{t+1} = \theta_t + \alpha \nabla_\theta J(\theta)$.

Metrics

① Average (state) value:

$$\bar{v}_\pi \triangleq \sum_{s\in S} \underbrace{d(s)}_{\text{distribution}} v_\pi(s) = E[v_\pi(S)] \quad , \quad (S \sim d)$$

vector form:

$$v_\pi = [\dots, v_\pi(s), \cdots]^T \in R^{|S|}$$

$$d = [\dots, d(s), \cdots]^T \in R^{|S|}$$

$$\Rightarrow \bar{v}_\pi = d^T v_\pi$$

$$\begin{cases} d \xrightarrow{\text{inde}} \pi \Rightarrow d_0, \ \bar{v}_\pi^0 \\ d \xrightarrow{\text{dep}} \pi \Rightarrow d_\pi, \ d_\pi^T P_\pi = d_\pi^T \end{cases}$$

② Average (one-step) reward:

$$\bar{r}_\pi \triangleq \sum_{s\in S} d_\pi(s) r_\pi(s) = E[r_\pi(S)]. \quad S \sim d_\pi$$

$$r_\pi(s) = \sum_{a\in A(s)} \pi(a|s) r(s,a)$$

$$r(s,a) = E[R|s,a] = \sum_r r p(r|s,a)$$

$$\overset{①}{\Rightarrow} \bar{r}_\pi = \sum_{s\in S} d_\pi(s) \sum_{a\in A(s)} \pi(a|s) \sum_r r p(r|s,a) = \sum_{s\in S}\sum_{a\in A(s)}\sum_r d_\pi(s)\pi(a|s) r p(r|s,a)$$

Another form, a trajectory $(R_{t+1}, R_{t+2}, \cdots)$

$$\overset{②}{\Rightarrow} \bar{r}_\pi = \lim_{n\to\infty} \frac{1}{n} E[R_{t+1} + R_{t+2} + \cdots + R_{t+n} | S_t = s_0]$$

$$= \lim_{n\to\infty} \frac{1}{n} E\left[\sum_{k=1}^{n} R_{t+k} | S_t = s_0\right]$$

$$= \lim \frac{1}{n} E\left[\sum^{n} R_{t+k}\right]$$

$$= \lim_{n\to\infty} \frac{1}{n} E\left[\sum_{k=1}^{n} R_{t+k}\right]$$

$$\gamma < 1 \Rightarrow \overline{r_\pi} = (1-\gamma)\overline{v_\pi}$$

Gradients

$$\boxed{\nabla_\theta J(\theta) = \sum_{s\in S} \eta(s) \sum_{a\in A} \nabla_\theta \pi(a|s,\theta) q_\pi(s,a)}$$

$$J(\theta): \quad \overline{v_\pi} \;/\; \overline{v_\pi^0} \;/\; \overline{r_\pi}$$
$$= \;:\; \quad = \;/\; \approx \;/\; \propto$$

$$\Rightarrow \begin{cases} \nabla_\theta \overline{r_\pi} \simeq \sum_s d_\pi(s) \sum_a \nabla_\theta \pi(a|s,\theta) q_\pi(s,a) \\ \nabla_\theta \overline{v_\pi} = \frac{1}{1-\gamma} \nabla_\theta \overline{r_\pi} \\ \nabla_\theta \overline{v_\pi^0} = \sum_{s\in S} \rho_\pi(s) \sum_{a\in A} \nabla_\theta \pi(a|s,\theta) q_\pi(s,a). \end{cases}$$

Transformation:

from

$$\nabla_\theta \ln \pi(a|s,\theta) = \frac{\nabla_\theta \pi(a|s,\theta)}{\pi(a|s,\theta)}$$

get

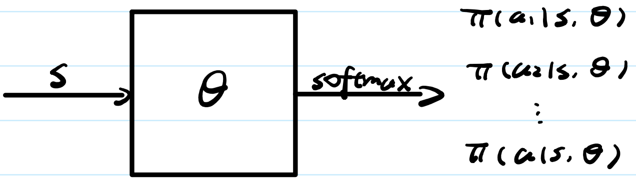
$$\nabla_\theta \pi(a|s,\theta) = \pi(a|s,\theta) \nabla_\theta \ln \pi(a|s,\theta)$$

$$\Rightarrow \nabla_\theta J = \sum_s d(s) \sum_a \pi(a|s,\theta) \nabla_\theta \ln \pi(a|s,\theta) q_\pi(s,a)$$

$$= E[\nabla_\theta \ln \pi(A|s,\theta) q_\pi(s,A)], \quad (S\sim d,\ A\sim\pi.)$$

Remarks:

$$\pi(a|s,\theta) > 0 \Rightarrow \text{softmax} \quad \boxed{\pi(a|s,\theta) = \frac{e^{h(s,a,\theta)}}{\sum_{a'\in A} e^{h(s,a',\theta)}}}$$



Policy Gradient + MC → REINFORCE

REINFORCE

REINFORCE

since

$$\theta_{t+1} = \theta_t + \alpha \left(\frac{q_t(s_t, a_t)}{\pi(a_t|s_t, \theta_t)} \right) \nabla_\theta \pi(a_t|s_t, \theta_t)$$

let

$$\beta_t = \frac{q_t(s_t, a_t)}{\pi(a_t|s_t, \theta_t)} \Rightarrow q_t \uparrow, \ \beta_t \uparrow \quad \text{exploitation}$$

$$\pi \downarrow, \ \beta_t \uparrow \quad \text{exploration}$$

Algo (REINFORCE)

Init: $\pi(a|s, \theta)$, $\gamma \in (0, 1)$, $\alpha > 0$.

kth iter:

episode (start s_0, follow $\pi(\theta_k)$): $\{s_0, a_0, r_1, \cdots s_{T-1}, a_{T-1}, s_T\}$

for $t = 0, 1, \cdots, T-1$, do:

① value update: $q_t(s_t, a_t) = \sum_{k=t+1}^{T} \gamma^{k-t-1} r_k$

② policy update: $\theta_{t+1} = \theta_t + \alpha \nabla_\theta \ln \pi(a_t|s_t, \theta_t) q_t(s_t, a_t)$

$\theta_k = \theta_T$.